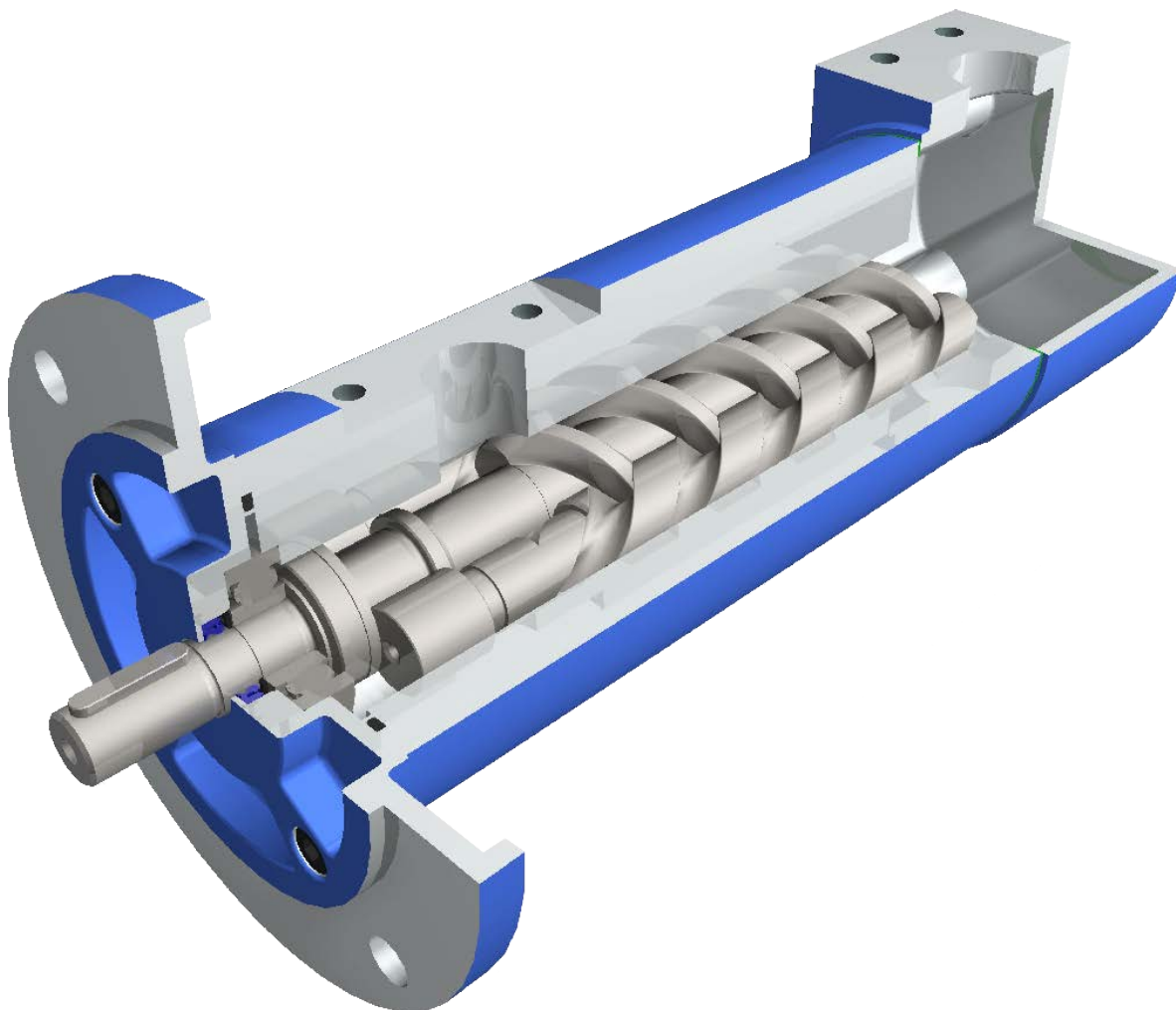




Pump & Motor Division

General Installation, Operation, Maintenance and
Troubleshooting Manual for Screw Pump Series



ENGINEERING YOUR SUCCESS.



WARNING — User Responsibility

FAILURE OR IMPROPER SELECTION OR IMPROPER USE OF THE PRODUCTS AND/OR SYSTEMS DESCRIBED HEREIN OR RELATED ITEMS CAN CAUSE DEATH, PERSONAL INJURY AND PROPERTY DAMAGE.

This document and other information from Parker Hannifin Corporation, its subsidiaries and authorized distributors provide product and/or system options for further investigation by users having technical expertise. It is important that you analyze all aspects of your application and review the information concerning the product or system in the current product catalog. Due to the variety of operating conditions and applications for these products or systems, the user, through its own analysis and testing, is solely responsible for making the the products and systems and assuring that all performance, safety and warning requirements of the application are met.

The products described herein, including without limitation, product features, specifications, designs, availability and pricing, are subject to change by Parker Hannifin Corporation and its subsidiaries at any time without notice.

Offer of Sale

The items described in this document are hereby offered for sale by Parker Hannifin Corporation, its subsidiaries or its authorized distributors. This offer and its acceptance are governed by the provisions stated in the "Offer of Sale".

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READ THIS ENTIRE PAGE BEFORE PROCEEDING

For the safety of personnel and to prevent damage to the equipment, the following nomenclature has been used in this manual:



DANGER: Failure to observe the precautions noted can result in severe injuries or death.



WARNING: Failure to observe the precautions noted can result in injury to personnel by accidental contact with the equipment or liquids. Protection should be provided by the user to prevent accidental contact.



CAUTION: Failure to observe the precautions noted can cause damage or failure of the equipment.

Non-compliance of safety instructions could affect safety for persons are identified by the following symbols:



Safety instructions where electrical safety is involved are identified by the following symbols:



Safety instructions which shall be considered for reasons of safe operation of the pump and/or protection of the pump itself are marked by the following text:

ATTENTION



WARNING: This manual, should be read thoroughly prior to pump installation, operation, maintenance or troubleshooting.



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1. General Information

The instructions found herein cover the general installation, operation, maintenance and troubleshooting of subject equipment.

NOTE: Individual contracts may have specific provisions that vary from this manual. For further detailed information and technical assistance to questions not answered by these manuals, please refer to the Pump & Motor Division at 704-730-2000.

This manual cannot possibly cover every situation connected with the installation, operation, maintenance and troubleshooting of the equipment supplied. Every effort was made to prepare the text of the manual so that engineering and design data was transformed into easily understood wording.

Parker must assume that the personnel assigned to operate and maintain the supplied equipment and apply this instruction manual, should have sufficient technical knowledge and experience to use safety and operational practices which may not be otherwise covered by this manual.

In applications, where equipment furnished by Parker is to become part of a process or other machinery, these instructions should be thoroughly reviewed to determine proper fit of the equipment into overall plant operational procedures.



WARNING: If installation, operation, and maintenance instructions are not correctly and strictly followed and observed, injury to personnel or serious damage to pump could result. Parker cannot accept responsibility for unsatisfactory performance or damage resulting from failure to comply with instructions.



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2. Transportation and Storage

Always protect the pump against taking in water and other contaminants. Store the pump in a clean, dry and relatively warm environment. Pumps are delivered with their internals oiled (unless specified otherwise by the customer order) and with protective covers in or over all openings. These covers should remain in place during the mounting and alignment procedures. The covers must be removed just prior to attaching system piping to pump. If pumps are to be stored in other than a clean, warm, or dry environment, or if they are to be stored for more than six months, follow the below information.

Storage Recommendations

Full Controlled Atmosphere - Requirements:

Uniform temperature between 5°C and 60°C throughout the room maintained at least 5.5°C above the dew point. Relative humidity of 50% or less. Dust accumulation should be minimal with no harmful fumes and preferably without ambient vibration. In case of presence of vibrations, vibration isolation pads must be used.

Partially Controlled Atmosphere - Requirements:

Temperature between -25°C and 60°C. The selected room should be as clean, dry as possible and preferably without ambient vibration. In case of presence of vibrations, vibration isolation pads must be used. If the ambient temperature exceeds 60°C, the space or structure should be enclosed but ventilated. The enclosing structure should be designed to protect the items from flying debris or other damage from high winds. If the items can be moved, it is suggested that the entire items be encased in a strong, transparent plastic bag. Before sealing the bag, a moisture indicator should be attached to the side of the items and several bags of silica gel desiccant placed inside the bag around the items. When the moisture indicator shows that the desiccant has lost its effectiveness, as by a change in color, the bag should be opened and the desiccant replaced with fresh material.

Inland dry climate - Conditions Usually Encountered:

Dust, sand, heat from the sun, and occasional rain or snow.

Requirements:

Temperature between -25°C and 60°C. If the ambient temperature exceeds 60°C, the space or structure should be enclosed but ventilated. Cover the items completely to exclude dirt, dust, moisture and other foreign materials. At a bare minimum, a heavy waterproof cover should be slipped over the items. After the items are covered as described, a shed of some sort should be erected to protect them from direct rain, snow and excessive direct sun heat. Proper wrapping will protect the items from blown sand and dirt. If the items can be moved, it is suggested that the entire items be encased in a strong, transparent plastic bag. Before sealing the bag, a moisture indicator should be attached to the side of the items and several bags of silica gel desiccant placed inside the bag around the items. When the moisture indicator shows that the desiccant has lost its effectiveness, as by a change in color, the bag should be opened and the desiccant replaced with fresh material.

Inland humid climate (Non Industrial Area) - Conditions Usually Encountered:

Dust, rain, snow, organic growth (fungus).

Requirements:

Temperature between -25°C and 60°C. If the ambient temperature exceeds 60°C, the space or structure should be enclosed but ventilated. Cover the items completely to exclude dirt, dust, moisture and other foreign materials. At a bare minimum, a heavy waterproof cover should be slipped over the items. After the items are covered as described, a shed of some sort should be erected to protect them from direct rain, snow and excessive direct sun heat. Proper wrapping will protect the items from blown sand and dirt. If the items can be moved, it is suggested that the entire items be encased in a strong, transparent plastic bag. Before sealing the bag, a moisture indicator should be attached to the side of the items and several bags of silica gel desiccant placed inside the bag around the items. When the moisture indicator shows that the desiccant has lost its effectiveness, as by a change in color, the bag should be opened and the desiccant replaced with fresh material.

Continued on Next Page



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2. Transportation and Storage (Continued)

Salty and Industrial Atmospheres - Conditions Usually Encountered:

Moisture impregnated with salts or other chemicals, salty dust rain, snow, fungus growth, fumes, coal and chemical soot.

Requirements

Temperature between -25°C and 60°C. If the ambient temperature exceeds 60°C, the space or structure should be enclosed but ventilated. Cover the items completely to exclude dirt, dust, moisture and other foreign materials. At a bare minimum, a heavy waterproof cover should be slipped over the items. After the items are covered as described, a shed of some sort should be erected to protect them from direct rain, snow and excessive direct sun heat. Proper wrapping will protect the items from blown sand and dirt. If the items can be moved, it is suggested that the entire items be encased in a strong, transparent plastic bag.

Before sealing the bag, a moisture indicator should be attached to the side of the items and several bags of silica gel desiccant placed inside the bag around the items. When the moisture indicator shows that the desiccant has lost its effectiveness, as by a change in color, the bag should be opened and the desiccant replaced with fresh material.

Rotating Parts



WARNING: Rotating parts can cut. Follow all safety rules.

All the rotating parts of the items should be manually rotated by hand for about 20 rounds. Follow the table below for the rotation frequency:

Storage Enviroments	Shaft Rotation Frequency
Full Controlled Atmosphere	Every Two (2) Months
Partially Controlled Atmosphere	Once Per Month
Inland Dry Climate	Once Per Month
Inland Humid Climate	Once Per Week
Salty and Industrial Atmosphere	Once Per Week



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3. Installation/Assembly

WARNING: Lifting a vertical pump/driver using straps or hooks attached to the pump or pump-to-driver bracket may be dangerous since the centre-of-gravity of the assembly may be higher than the points of attachment. Take precautions to prevent slippage of slings and hooks. Always use properly rated lifting devices.

3.1. Tools

The procedures described in this manual require common mechanics hand tools, dial indicators for alignment and suitable lifting devices such as slings, straps, spreader bars, etc.

3.2. Lifting of Pump and Pump/Driver Assemblies

All pumps and pump/driver assemblies should be lifted with appropriate devices securely attached to the whole unit. Ensure unit's centre-of-gravity is located between lifting points (see Figure 1). This will avoid tipping of pump or pump/driver assembly. Spreader bars should be used as necessary to ensure load is properly distributed and lifting straps do not damage equipment.

Some pumps and pump/driver assemblies have designated lifting points that are shown on their outline drawings.

DANGER/ATTENTION: Lifting a vertical pump/driver using straps or hooks attached to the pump or pump-to-driver bracket may be dangerous since the centre-of-gravity of the assembly may be higher than the points of attachment. Take precautions to prevent slippage of slings and hooks. Always use properly rated lifting devices.

3.3. Installation of Pump Assembly

To ensure adequate flow of liquid to pump's inlet port, place pump near liquid source and preferably place pump centre line below liquid surface. Use short, straight inlet lines.

A dry, clean, well-lit and well-ventilated site should be selected for installing the pump assembly.

Sufficient open space should be provided around pump rotor and/or gear housing to permit routine visual inspection, on-site service and maintenance, and pump replacement. For installation and servicing of large pump units, ample overhead clearance should be provided to allow for lifting device manoeuvring.

In case of standard pump configuration, the suction connection can be rotated 90°, 180° or 270°.

The steps for the rotations are:

1. Remove the four screws.
2. Rotate the suction connection to desired position
3. Tighten the four screws in. Make sure that the sealing is properly located in its seat.

Tighten the screws according to the specification:
M 8 = 25 Nm, M 10 = 50 Nm, M 12 = 80 Nm.

4. Connect the pressure and suction lines and remove the protective stoppers.

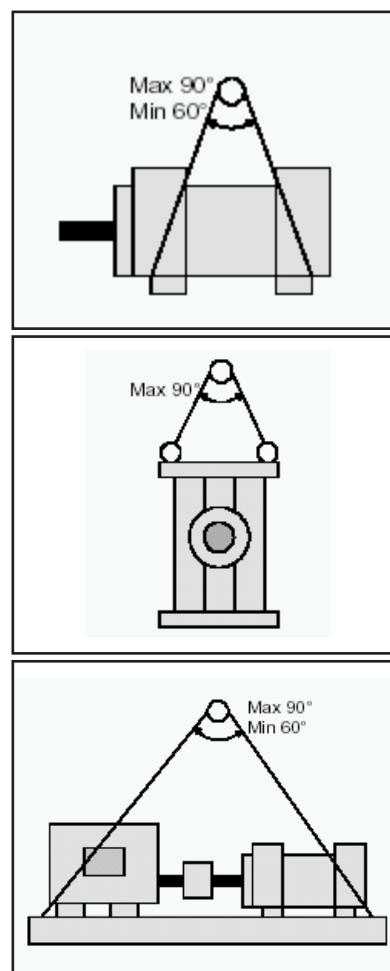


Figure 1 - Lifting Pumps and Pump/Driver Assemblies

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3. Installation/Assembly (Continued)

3.4. Foundations And Baseplates

Foundations and base plates must be designed and installed so pump and driver alignment can be maintained at all times. Be sure base plates are level and rest on smooth flat surfaces. Small pumps may be mounted on base plates or directly to existing floors that meet the criteria of foundations. Larger pumps and/or drivers must be mounted to base plates and foundations. It is recommended that pumps and their drivers be mounted on common base plates.

3.5. Mounting of Foot Mounted Pumps and Drivers

Some pumps are shipped on base plates without drivers. For these units, install and tighten each coupling half on driver and pump shafts. Place driver on base plate and set proper distance between shafts and coupling hubs (See **Figure 2**). Locate driver so pump and driver shafts are in axial alignment. See section 3.6 on Alignment.

For pumps driven through a separate gearbox or other device, first align device relative to pump, and then align driver relative to device.

See section 3.6.5 on Belts and Sheaves (page 8) for belt-driven pumps. On horizontal pump/driver assemblies, shaft couplings are often shipped disassembled to prevent coupling damage during shipping and handling.

When not supplied by the manufacturer, coupling, shaft and/or belt guards conforming to ANSI B15.1 should be installed for personnel protection during pump operation. Final alignment of pump and driver should take place after unit is secured to foundation. If base plate is to be grouted, this should be completed before final alignment.

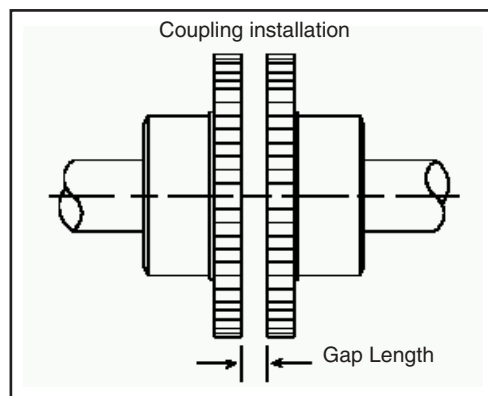


Figure 2 - Coupling Gap Measurements

NOTE: Grouting is recommended to prevent lateral shifting of base plate, not to take up irregularities in the foundation. For installations requiring grouting, a base plate designed specifically for this purpose is needed.



WARNING: Install guards over couplings and shafts to protect personnel from accidental contact with rotating couplings, belts, sheaves, chains, shafts and/or keyways.

3.6. Alignment

3.6.1. General

All pump and driver assemblies must be aligned after site installation and at regular maintenance intervals. This applies to factory-mounted units (new or rebuilt) because factory alignment is often disturbed during shipping and handling. Flexible couplings shall be used to connect pump to its driver (unless otherwise specified by Parker).

The objective of any aligning procedure is to align shafts (not align coupling hubs) by using methods that cancel out any surface irregularities, shaft-end float, and eccentricity.

At operating temperatures above 65°C (175°F), pumps require "hot alignment" after pump and driver reach normal operating temperatures. Also, re-check final alignment after all piping is connected to pump.



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3. Installation/Assembly (Continued)

3.6.2. Flexible Shaft Couplings

Flexible couplings are intended to provide a mechanically flexible connection for two aligned shaft-ends. Flexible couplings are not intended to compensate for major angular or parallel shaft misalignment. The allowable misalignment varies with the type of coupling. Any improvement in alignment beyond coupling manufacturer's minimum specification will extend pump, mechanical seal or packing, coupling, and driver service life by reducing bearing loads and wear.



CAUTION/ATTENTION: Flexible couplings are NOT intended to permit significant shaft misalignment. Proper alignment must be established/maintained to obtain proper operation and maximum life. Pump alignment requirements are nearly always more strict than coupling alignment requirements. Regardless of coupling manufacturer's stated limits, pump-to-driver shaft alignment must be per pump's alignment requirement. Be sure all coupling set-screws and bolts are tight and coupling gap is properly set.

3.6.3. Aligning Foot Mounted Pumps

To install foot mounted pumps (see **Figure 3**), perform the following:

1. Install pump and driver onto base plate after installing appropriate coupling halves on pump and driver shafts.
2. Perform alignment of pump and driver shafts using dial indicators. Acceptable alignment has been attained when FIM (Full Indicator Movement) is less than or equal to 0.1mm (0.005 inch) for face (angularity) and rim (parallelism) readings at or near coupling outer diameter while rotating both shafts together one full turn (360°).

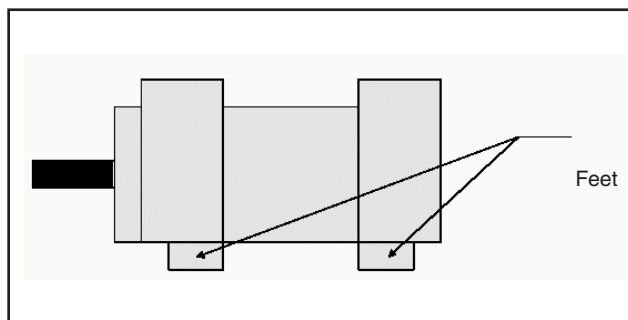


Figure 3 - Foot Mounted Pump

3.6.4. Aligning Flange Mounted Foot Mounted Pumps

Shaft alignment requirements for flange mounted pumps (see **Figure 4**) are the same as for foot mounted pumps. That is, shafts must be aligned within 0.1mm (0.005 inch) FIM (Full Indicator Movement) for face (angularity) and rim (parallelism) at or near coupling outer diameter while rotating both shafts together one full turn (360°).

When a pump and driver are both flange mounted to a bracket, DO NOT assume bracket will automatically align shafts to the above requirements. Brackets must be designed to obtain/maintain required alignment as well as to support pump weight plus any (small) residual piping forces without distorting. If at all possible, bracket design should include adequate room to check shaft alignment with dial indicators with both pump and motor mounted onto bracket. If this is not possible, align bracket to drive shaft (see **Figure 5**), then attach pump to bracket (assumes pump fits snugly into its mounting bore in the bracket).

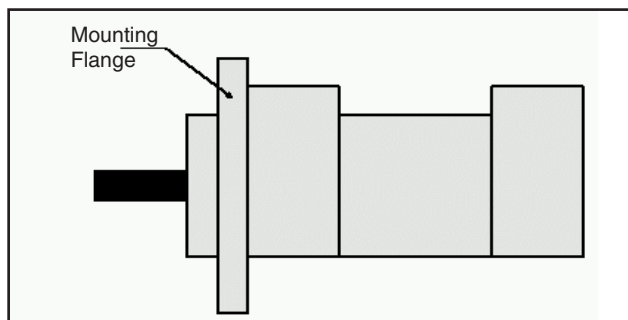


Figure 4 - Flange Mounted Pump

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3. Installation/Assembly (Continued)

3.6.4. Aligning Flange Mounted Foot Mounted Pumps (Continued)

After pump-bracket-driver is installed into system and after piping is connected to pump, shaft alignment should be re-checked and adjusted, if necessary. When a right-angle foot bracket is used, mount pump onto bracket and tighten pump-to-bracket mounting bolts. At this point, bracket base assumes the functionality of pump feet. Continue with aligning procedure as if pump were feet mounted. See section 3.6.3 on Aligning Foot Mounted Pumps (page 6).



CAUTION/ATTENTION: Be sure all coupling set screws and bolts are tight and coupling gap is properly set. Install safety shield(s) or plate(s) over bracket opening(s).

1. Place a dial indicator on the coupling as shown in the picture (see **Figure 5a**).
2. Check the vertical alignment: set the dial indicator to the zero mark on the top of the coupling, rotate the motor shaft of 180 degrees and read the misalignment, MAX 0.1 mm (see **Figure 5b**).
3. Check the horizontal alignment: set the dial indicator on one side of the coupling, rotate the motor shaft of 180 degrees and read the misalignment, MAX 0.1 mm (see **Figure 5c**).

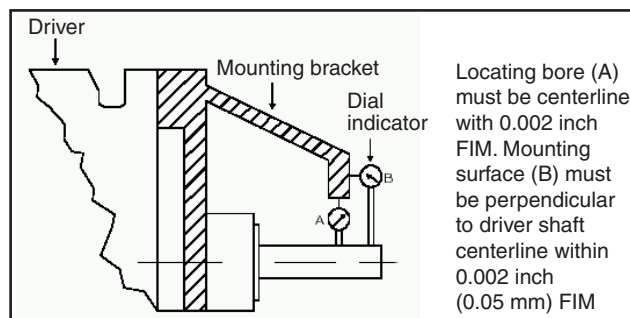


Figure 5 - Alignment of Flange Mounted Pump

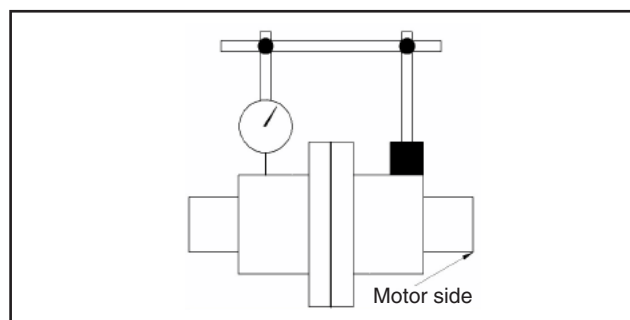


Figure 5a

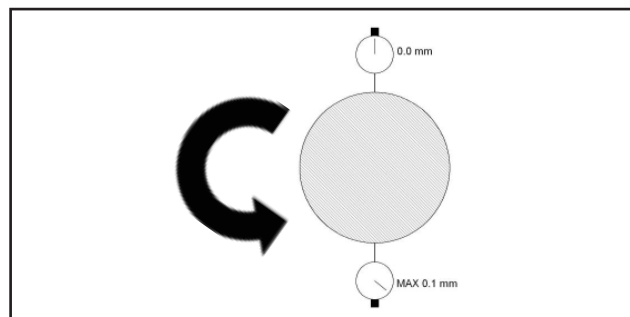


Figure 5b

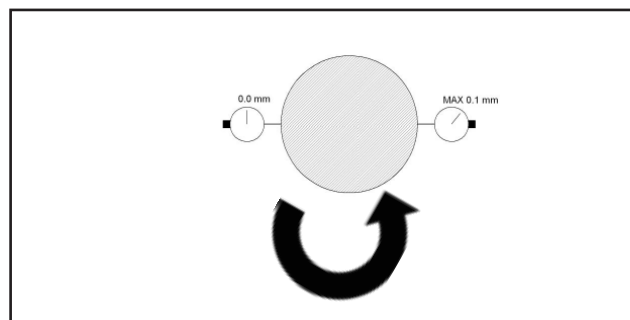


Figure 5c



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3. Installation/Assembly (Continued)

3.6.5. Belts and Sheaves

It is only acceptable to belt drive Parker pumps that are specifically designed for this purpose. It is generally not acceptable to belt drive pumps with ratings in excess of 40 bar (600 psi) differential pressure. Contact Parker if not sure a particular pump can be belt driven.

Belts and sheaves must be properly selected aligned and tensioned to minimize belt wear, eliminate possibility of belt turnover in sheave grooves, and avoid excessive side load on pump shaft. Adjustable slide rails mounted under driver are recommended for proper belt tensioning.

Check belt tension frequently during first 24 to 48 hours of run-in operation. Follow belt drive manufacturer's recommendations for alignment of sheaves and belt-tension settings.



CAUTION/ATTENTION: Loose, slipping belts will squeal and cause overheating of sheaves leading to reduced belt life. Excessively tightened belts will result in reduced belt and bearing life and possible bearing or shaft failure.

3.6.6. Hollow Shaft

Close all cut-off cocks and remove protection for outlet and inlet. To facilitate venting, place the outlet higher than inlet.

Proceed as follow:

- Check the motor: verify the perpendicularity of the flange and the motor shaft: 0,05 max allowed.
- The use of motor IP65 is suggested.
- Warranty is voided if motor is outside the tolerances.
- Put the motor in **vertical position**, as per drawing (see **Figure 6**).
- The pump has to **entry free** on the shaft of electric motor.
- **Do not force**. If necessary remove and polish the key of the shaft of electric motor.
- After you have tightened the four screws, verify that the pump-motor group **turn free by rotating the motor fan**. If not the shafts may be misaligned.
- In case of replacement, and the motor cannot be disassembled, always verify the free smooth rotation of the motor fan. **IF THE ROTATION OF THE MOTOR FAN IS NOT SMOOTH, DO NOT ACTIVE THE MOTOR.**



CAUTION/ATTENTION: To reduce possible FRETTING corrossions, please use appropriate greases to lubricate the motor shaft.

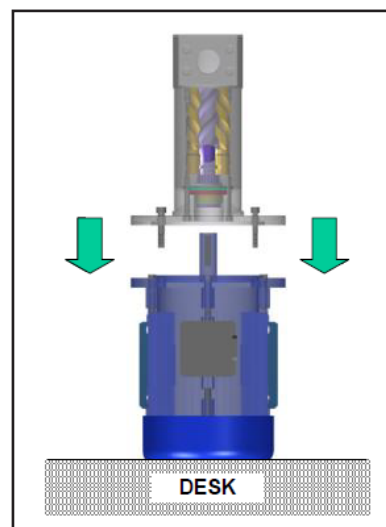


Figure 6 - Hollow shaft mounting

NOTES:

- FRETTING: To reduce the corrosion due to fretting effect we recommend to greases the motor shaft with dedicated products (samples: lubricants based on MoS2, Loctite® 8008, Molykote® G-n plus, Turmopast® MA2).
- FRETTING: To reduce the corrosion due to fretting effect, we recommend to check the electric motor ground connection and to check that the shaft residual currents are within the norms.
- LEACKAGE PREVENTION: In case of wear of shaft seal to avoid leakage, all pump flanges with hallow shaft have a threaded 1/4" GAS thread that can be used for drainage connection to the tank.



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3. Installation/Assembly (Continued)

3.6.7. Steel Lamina Coupling – Vertical Installation

The coupling has to be assembled to the shaft with its calibrated spacer, the calibrated spacer is already installed into the joint by Parker (see **Figure 7a**).

Lift the pump by eyebolts and fix it on its base (see **Figure 7b**). Then remove eyebolts.

Install the bell-housing with the spacer by four screws (see **Figure 7c**).

If necessary align the hexagon socket set screws of the coupling (motor side only) with the hole 1 of the bell housing to make easy the installation of the motor shaft (see **Figure 7d**).

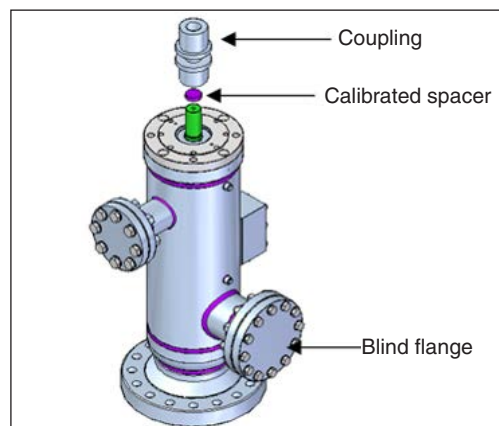


Figure 7a

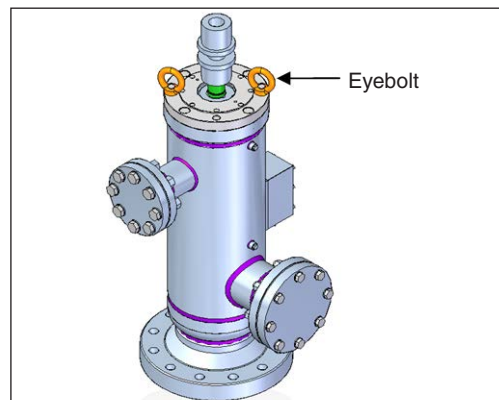


Figure 7b

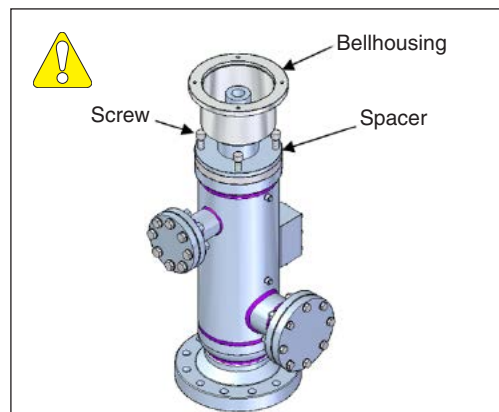


Figure 7c

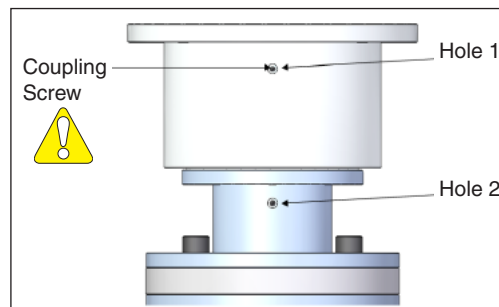


Figure 7d



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3. Installation/Assembly (Continued)

3.6.6. Steel Lamina Coupling – Vertical Installation (Continued)

Hole 2 is intended for inspection only. Prepare the motor in the right position above the pump and keep it suspended by a proper lifting system (see **Figure 7e**).

Before motor installation you have to pre-heat the half joint coupling by an air heater up to 100°C (see **Figure 7f**).

Remove the blind flanges (see **Figure 7f**) to the suction and the pressure port and connect suction and pressure port with piping.

ATTENTION: Install the motor immediately (in order to minimize joint cooling) fixing it to the bell housing by four bolts. Tighten the hexagon socket set screws of the coupling, motor side only

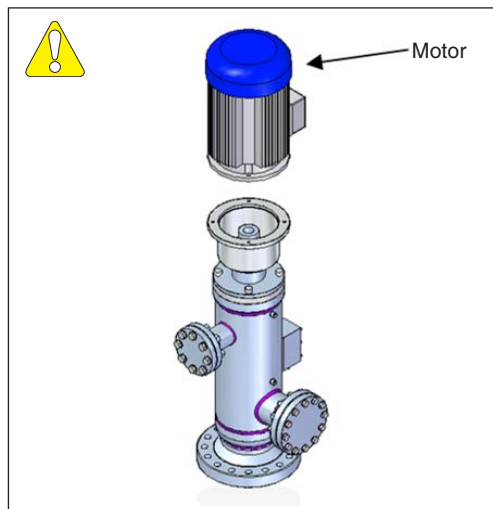


Figure 7e

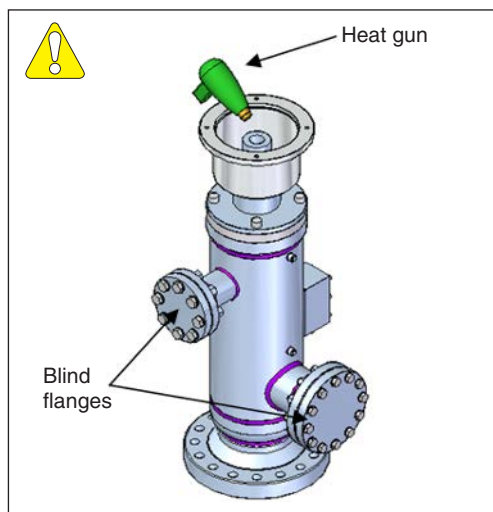


Figure 7f

3.7. Piping and Valves

3.7.1. General

Piping connected to pump **MUST** be independently supported and not allowed to impose strains on pump casing including allowing for expansion and contraction due to pressure and temperature changes.

To prevent foaming and air entrainment, all return lines in re-circulating systems should end well below liquid surface in reservoir. Bypass liquid from relief pressure and flow control valves should be returned to source (tank, reservoir, etc.), **NOT** to pump inlet line.

Shut-off valves should be installed in both the suction and discharge lines so pump can be hydraulically isolated for service or removal. All new piping should be flushed clean before connecting to pump.

 **CAUTION/ATTENTION:** Pipe strain will distort a pump. This could lead to pump and piping malfunction or failure. Return lines piped back to pump can cause excessive temperature rise at pump which could result in catastrophic pump failure.

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3. Installation/Assembly (Continued)

3.7.2. Relief Valve

Use relief valves to protect pumps from overpressure. They need to be connected to pump discharge lines as close to pumps as possible and with no other valves between pumps and relief valves. Relief valve settings should be set as low as practical.

DO NOT set relief valve higher than maximum pressure rating of pump, including pressure accumulation at 100% bypass. Relief valve return lines should NOT be piped into pump inlet lines because they can produce a loop that will overheat pump (see **Figure 7g**).



DANGER: The Parker pump is a positive displacement type. It will deliver (or attempt to deliver) flow regardless of back-pressure on unit. Failure to provide pump overpressure protection can cause pump or driver malfunction and/or rupture of pump and/or piping.

Some low pressure pump models include built-in safety relief valves. They are intended only for emergency operation, NOT for system control. Extended operation of relief valves in these pumps could lead to pump damage or failure.

3.7.3. Suction Line

The suction line should be designed so pump inlet pressure, measured at pump inlet flange, is greater than or equal to the minimum required pump inlet pressure (also referred to as Net Positive Inlet Pressure Required or (NPIPR). Velocity in suction line should be kept within 0,5÷1,2 m/s. Suction line length should be as short as possible and equal to or larger than pump's inlet size. All joints in suction line must be tight and sealed. If pump cannot be located below liquid level in reservoir, it is necessary either to position the suction line as in **Figure 8a** or install a foot valve so liquid cannot drain from pump while it is shut down (see **Figure 8b**). When pump is mounted vertically with drive shaft upward, or mounted horizontally with inlet port opening other than facing upward, a foot valve or liquid trap should be installed in suction line to prevent draining. The suction line should be filled before pump start-up.



CAUTION/ATTENTION: DO NOT operate the pump without liquid or under severe cavitation.

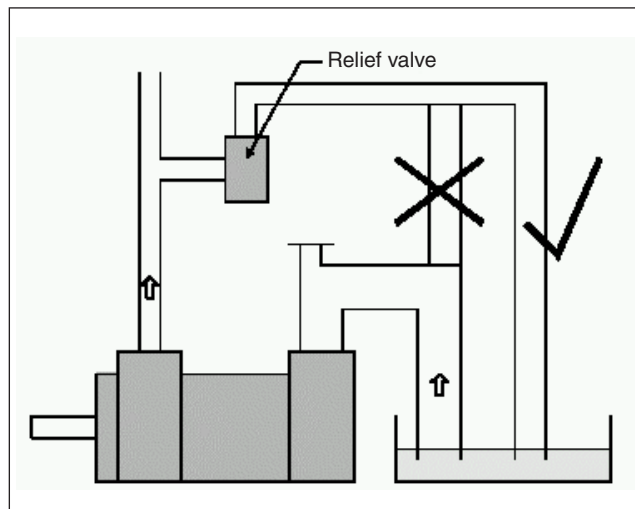


Figure 7g - Proper Relief Valve Return Line Arrangement

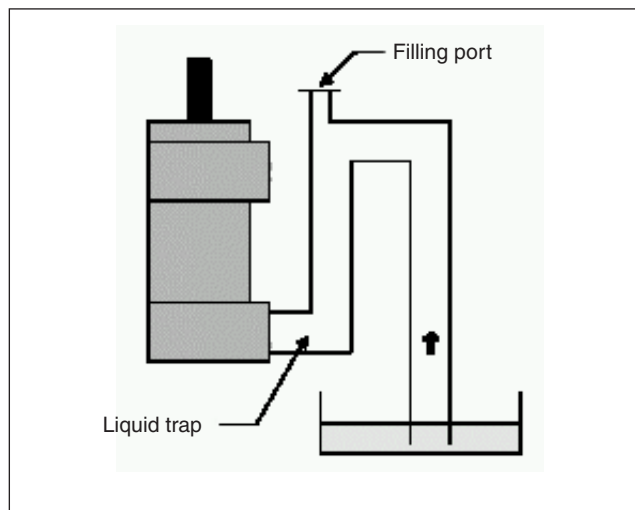


Figure 8a - Fluid Trap & Foot Valve Arrangements for Vertical Pumps

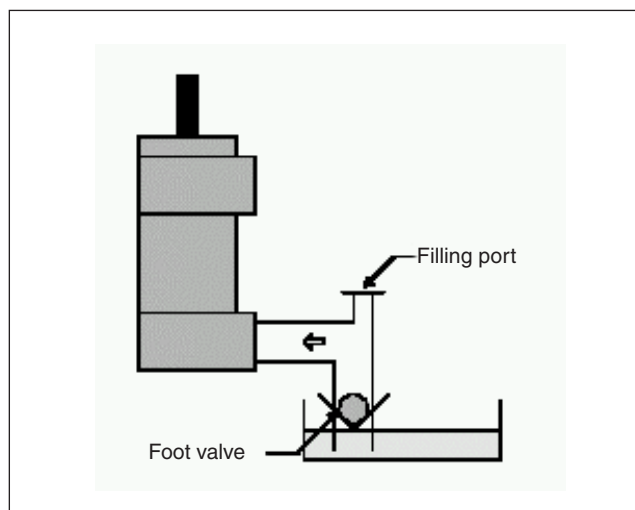


Figure 8a - Fluid Trap & Foot Valve Arrangements for Vertical Pumps



WARNING: This product can expose you to chemicals including lead or DEHP which are known to the state of California to cause cancer, birth defects, and other reproductive harm. www.p65warnings.ca.gov

3. Installation/Assembly (Continued)

3.7.4. Suction Strainer /Filter

Pump life is related to liquid cleanliness. Suction strainers or filters should be installed in all systems to prevent entry of large contaminants into pump (see **Figure 9**).

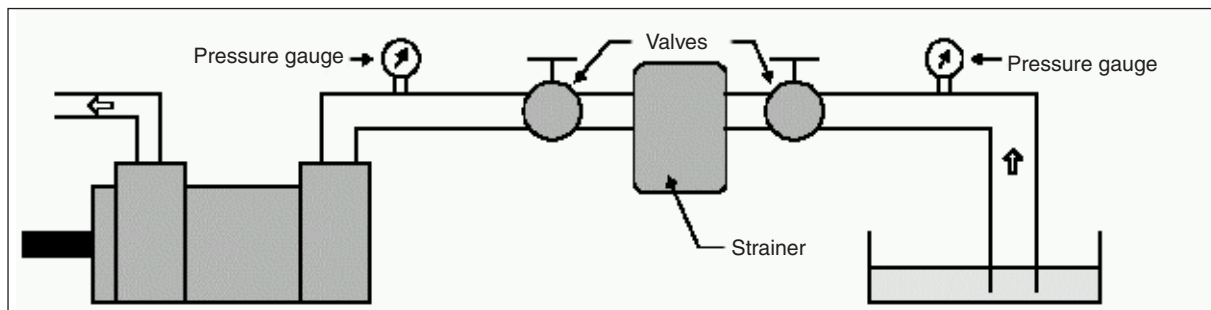


Figure 9 - Ideal Strainer Arrangement

The purpose of a suction strainer or filter is for basic protection of internal pumping elements. It should be installed immediately ahead of inlet port. This location should provide for easy cleaning or replacement of strainer element. Appropriate gauges or instrumentation should be provided to monitor pump pressure.

Pressure drop across a dirty strainer must not allow inlet pressure to fall below NPIPR. The pressure drop across the strainer should preferably not exceed 0.1 bar at max. flow rate and normal operating viscosity.

General guidelines for strainer sizing are as follows:

- When pumping relatively clean viscous liquids (over 1000 cSt), use 10 to 12 mesh screens or those with about 1.5mm (1/16 inch) openings.
- When pumping relatively clean light liquids such as distillate fuels, hydraulic oil and light lube oils, use suction strainers of 100 to 200 mesh.
- When pumping heavy crude oils, use 5 to 6 mesh strainer screens or those with or about 3mm (1/8 inch) openings.
- When pumping relatively clean distillate fuels in high pressure fuel supply systems, use 25 micron “absolute” filters for three screw pumps and 10 micron “absolute” filters for gear pumps.
- Make sure size/capacity of strainer or filter is adequate to prevent having to clean or replace elements too frequently.



CAUTION/ATTENTION: Before connecting pump to system, all system piping must be thoroughly flushed to remove debris which accumulates during fabrication, storage, and installation. Parker pumps should not be used for flushing. One large, hard particle may cause internal damage, possibly requiring a pump overhaul. Pay particular attention to suction line between suction strainer and pump to be sure it is clean.



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3. Installation/Assembly (Continued)

3.7.5. System Filtration

In systems that re-circulate the pumped liquid, downstream (pressure and/or return side) filtration should be installed. Downstream filters may also be required to protect components such as servo valves in hydraulic systems or high-pressure fuel nozzles and flow dividers in fuel oil supply systems for gas turbines.

The system's most contamination-sensitive component determines its liquid cleanliness requirement. For optimum Parker pump life when running on fuel oil, light lube oil, hydraulic oil and other relatively low viscosity (thin) liquids, a high efficiency 10 micron "absolute" or finer filter is recommended in accordance with NAS 1638/10 or ISO DIN 4406-19/16. This same filter rating is recommended for pumps running at extreme operating conditions and/or in harsh environments. For pumps running on relatively clean, more viscous (thicker) liquids, filter ratings as high as 25 micron "nominal" may be acceptable as long as operating conditions and the operating environment are moderate.

Parker should be contacted for filtration requirements for pumps running on very low viscosity (water thin) and low lubrication as well as for those with an unusually large quantity of contaminants.

The system builder determines filter size (dirt holding capacity) by the amount and size of contamination expected to be produced by system and other external contamination sources, by allowable pressure drop across filter and by acceptable frequency for cleaning/replacing filter elements.

3.7.6. Outlet Piping

In general, outlet piping should be sized to accommodate the pump's flow rate while minimizing pipe friction losses. Velocity in pressure line should be kept within $1\div3$ m/s. It should also be designed to prevent gas and air pockets. Piping downstream of pump should include a vent at highest point in system to allow air to escape during priming.

3.8. Shaft Packing and Seal Leakage

The pump should be installed so any leakage from shaft packing or shaft seal does not become a hazard. Packing leakage should be about 8 to 10 drops per minute. A small amount of liquid may also leak from mechanical or lip seals (usually less than or equal to 10 drops per hour).



WARNING/DANGER: If not appropriately collected, packing leakage may make floor slippery or expose personnel to hazardous fluids. Provisions should be made to collect leakage from packing or shaft seals.

3.9. Quenched Shaft Seals

Some pumps include quenched mechanical shaft seals. For these pumps, a low pressure stream of steam, nitrogen, or clean water is supplied from an external source to atmospheric side of seal faces. Quenching is used in selected seal applications to:

- Heat or cool seal area.
- Prevent build-up of coke formations by excluding oxygen.
- Flush away undesirable material build-up around dynamic seal components.
- When quenched mechanical seals are part of pump assembly, an appropriate quenching stream must be supplied by user.

NOTE: Refer to pump or pump/driver outline drawing for quench connection size and port locations.



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3. Installation/Assembly (Continued)

3.10. Gauges

Pressure and temperature gauges are recommended for monitoring the pump's operating conditions. These gauges should be easily readable and placed as close as possible to pump's inlet and outlet flanges (see **Figure 10a**).

3.11. Idealized Installation for Pumps Located Above Liquid Level

Figures 10a and 10b are compilations of Figures 7g, 8 and 9 ([pages 11-12](#)) showing good-practice installation schemes for pumps located above the liquid reservoir in systems that re-circulate the pumped liquid.

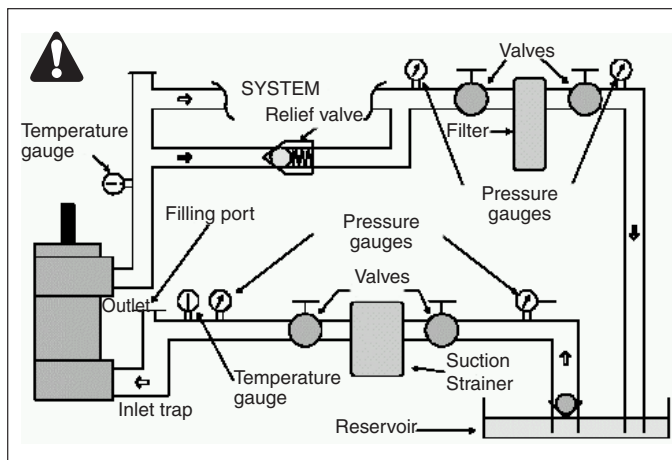


Figure 10a - Vertical Mounted Pump

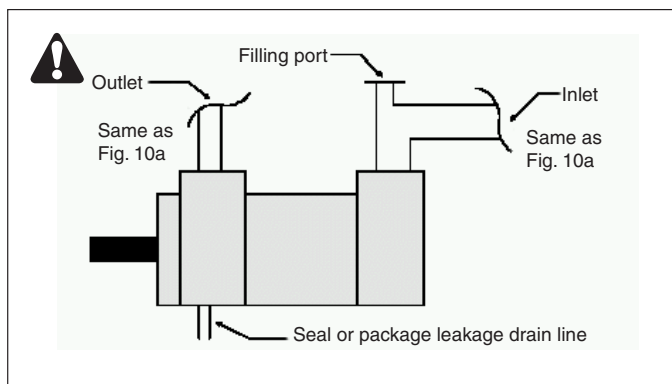


Figure 10b - Horizontal Mounted Pump



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4. Start Up, Operation & Shut Down



CAUTION/ATTENTION: Operation conditions, such as speed, liquid viscosity, temperature, inlet pressure, discharge pressure, filtration, duty cycle, drive type, mounting, etc., are interrelated. Due to these variable conditions, specific application limits may be different from pump's operating and structural limits. This equipment must not be operated without verifying system's operating requirements are within the pump's capabilities.



DANGER: Make sure all power equipment is disconnected and locked-out before proceeding.

4.1. Electrical Connections

Verify electrical requirements for driver match electrical supply with respect to voltage, number of phases and terminal connections. Also, check that driver has been wired to rotate in correct direction.

4.2. Rotation

Before connecting couplings, check pump rotation to be sure it matches rotation of driver. When coupling is connected and shafts are correctly aligned, pump should turn freely by hand. Rotation direction is indicated by an arrow cast on casing or by an attached plate showing a rotation direction arrow (see **Figure 11**).

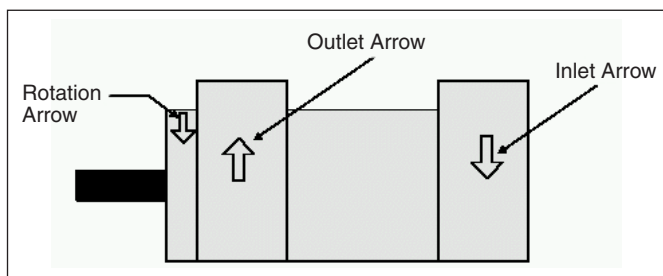


Figure 11 - Rotation Arrow



CAUTION/ATTENTION: Operating pump in the reverse direction may cause pump damage. Make sure rotation direction is not confused with inlet or outlet flow direction arrows.

4.3. Hydrostatic Testing the System

Before any system is hydrostatically tested, pump must be removed or isolated.



CAUTION/ATTENTION: To prevent damage to pump, it is necessary to remove or isolate it from the system prior to starting hydrostatic testing.

4.4. Protective Devices

4.4.1. General

Automatic shutdowns, emergency switches, and similar controls should be part of pumping system. They are generally supplied by system supplier or user.

4.4.2. Covers and Guards

Before start-up, ensure all protective-covers and guards are in place.



WARNING/DANGER: To protect personnel from accidental contact with rotating couplings, sheaves, belts, shafts keys, keyways, etc., install the following covers or guards over: Bracket openings on flange mounted pumps. Couplings and shafts on foot mounted pumps. Sheaves, gears, chains, belts or other type drives.



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4. Start Up, Operation & Shut Down (Continued)

4.4.3. Valves

Check all valves, especially those that are manually operated, to be sure they are in the proper position. Check that there is no possibility of starting pump with a blocked suction or discharge line.



WARNING: Starting a pump with discharge line blocked and without adequate relief protection will cause catastrophic pump failure and possible injury to personnel.

4.5. Intermediate Drive Lubrication

Some Parker pump units include intermediate gearboxes or other devices between pump and driver. When these devices are present, lubrication is required. Add lubricant to specified level per device manufacturer's recommendations before start-up.

4.6. Quenched Shaft Seals

When quenching fluid is hot water or steam, apply to seal at least 30 minutes prior to pump start-up to ensure seal area is thoroughly heated. When steam is used, it should be saturated at about 4 to 7 psi gauge. When quench fluid is ambient temperature nitrogen, it can be applied just prior to pump start-up.

4.7. Pumped Liquids

NEVER operate a pump with water. The pump is designed for liquids having general characteristics of oil. In closed or re-circulating systems, check liquid level in tank before and after start-up to be sure it is within operating limits. If initial liquid level is low, or if it drops as system fills during start-up or pumping operations, add sufficient clean liquid to tank to bring liquid to its normal operating level. Only use liquid recommended or approved for use with the equipment. Regular checks should be made on the condition of the liquid. In closed systems, follow supplier's recommendations for maintaining liquid and establishing when liquid is to be changed. Be sure temperature is controlled so liquid cannot fall below its minimum allowable viscosity which occurs at its maximum operating temperature. Also, ensure that maximum viscosity at cold startup does not cause pump inlet pressure to fall below its minimum required value.



CAUTION/ATTENTION: NEVER operate a pump without liquid in it! Operate only on liquids approved for use with pump.



WARNING: If not appropriately collected, packing or seal leakage may make floor slippery and/or expose personnel to hazardous fluids.

4.7.1. High Viscosity

High Viscosity and high rotation speed may cause the pump to cavitate. Consequence of this may with time damage the pumps inner part and deteriorate the efficiency of the pump if not the overall pump behaviour. Special finishing of the internal screws may prevent the cavitations due to high viscosity with a loss of overall efficiency.

When application requires fluid viscosity higher than 68 cSt special option have to be selected when ordering the products. Ask Parker for support when selecting a product for high viscosity. Use the table on [page 30](#) to guide your application description.



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4. Start Up, Operation & Shut Down (Continued)**4.7.2. Air Emulsions**

In gear box or turbine lubrication applications air emulsion may not be avoided. Air Emulsion may cause the pump to cavitate. Consequence of this may with time damage the pumps inner part and deteriorate the efficiency of the pump if not the overall pump behaviour. Special finishing of the internal screws may prevent the cavitations due to high viscosity with a loss of overall efficiency.

When in the application there is air emulsion fluids, special options have to be selected when ordering the screw pump. Ask Parker for support when selecting a product to operate in high viscosity fluids. Use the table [page 30](#) to guide your application description.

4.8. Priming**4.8.1 General**

Prime pump before initial start-up by pouring some of liquid to be pumped into fill point in system or directly into pump suction port. Rotate pump slowly by hand until rotors or gears (pumping elements) are wet and suction line is as full of liquid as possible (see **Figure 12a**). Also, fill mechanical seal chamber with liquid to ensure seal does not start dry.



CAUTION/ATTENTION: Dry-starting a pump is likely to cause damage to pumping elements and shaft seal.

4.8.2. SMAPI Pumps

Open suction line to fill-in pump case suction side giving vent to case suction chamber with proper vent cap.

Open pressure line to fill-in pump case pressure side using with proper vent cap (see **Figure 12b**).

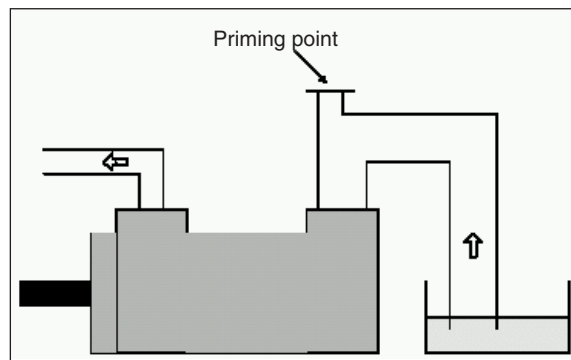


Figure 12a

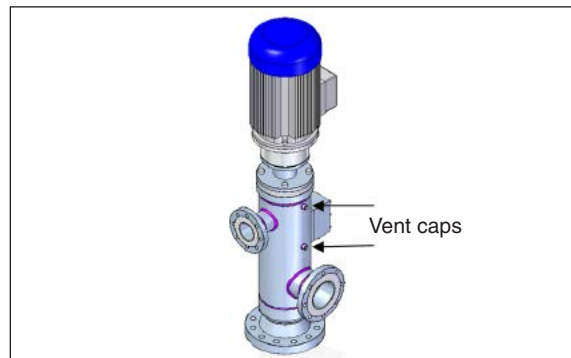


Figure 12b

4.9. Start-Up

It is suggested that the driver be started and immediately stopped (jogged) three or four times in order to verify proper pump rotation and to ensure pump is filled with liquid. Open bleed port at high point in system and vent trapped air until a solid stream of liquid emerges (where practical). For SMAPI pumps use proper vent cap available onto pump case to assure any air is into pump, piping and system.

In the setting of control software parameters consider that this type of pump must avoid the starting against a direct pressure.

Let the motor arrive to nominal RPM without close the pressure setting valve, the ideal time is 5 seconds, at least 3 seconds.

When pump is running, check for unusual noise or vibration. Investigate any abnormalities.

Check inlet and outlet gauges to see if pump is operating within its ratings. Generally, differential pressure across pump should be at least 1.5 bar (25 psi) to ensure proper pump operation.



CAUTION/ATTENTION: Before starting, make sure the prime mover is locked out and can not be started accidentally. Precautions must be taken when venting air in system using hazardous liquids. Provide hearing protection whenever high noise levels are expected from system components and/or environment. If operating temperatures exceed 60°C (140° F), measures should be taken to avoid skin contact.



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4. Start Up, Operation & Shut Down (Continued)

4.10. Shaft Packing (Stuffing Box) Leakage

Pumps with packing-type seals must be checked to ensure packing gland is not too tight. Excessive gland pressure on packing will cause a scored shaft, overheating and rapid breakdown of packing. Keep gland nuts only finger tight. After new packing has been installed, gland nuts should be tightened evenly but only tight enough to seat packing rings properly. Then, loosen gland nuts and re-tighten finger tight. The final adjustment should allow a leakage of approximately ten drops per minute while pump is operating. This leakage is necessary to lubricate the packing. Provide a place for safe draining and disposal of this leakage.



WARNING: If not appropriately collected, packing leakage may make floor slippery and/or expose personnel to hazardous fluids.

4.11. RP Valve Regulation

RP valve (Adjustable relief valve) is just a safety valve not a regulation valve.

Depending on pump type and size RP valve can be different (see **Figure 13a, b, c**)

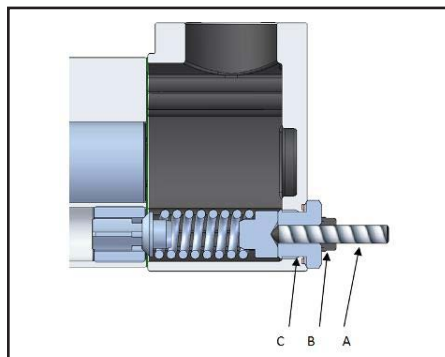


Figure 13a

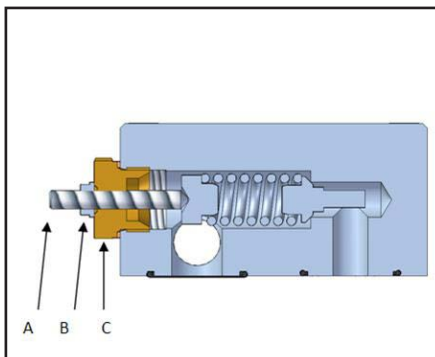


Figure 13b

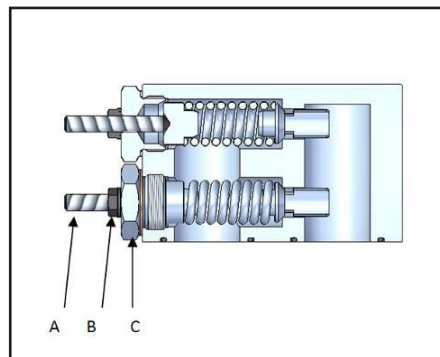


Figure 13c

4.11. RP Valve Regulation

To adjust relief pressure un-tighten nut B than:

- Tighten screw A to increase relief pressure
- Un-tighten screw A to decrease relief pressure
- Check that C nut is completely screwed
- Tighten nut B

For multiples valve (see **Figure 13c**) screws A must set at the same length. For SFO pump Series adjust screw A (see **Figure 13d**)

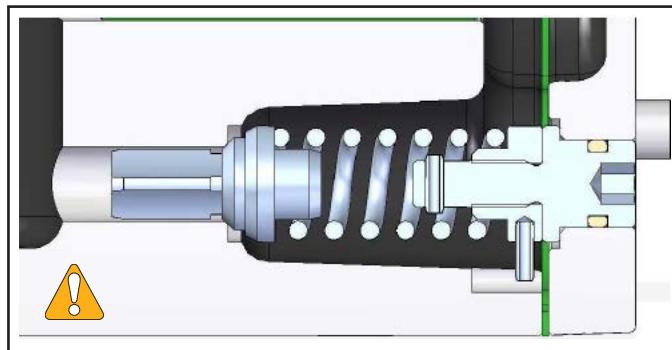


Figure 13d



DANGER: A similar sequence preserve the pump for a longer life. In general avoid in any automatic / manual start-up procedure all conditions of start / stop with pipeline in pressure.

In operations or maintenance process avoid a pipeline pressure condition that let the pump run with the internal bypass open for more than 3 seconds (3"). If, for special reason, it cannot be achieved, please follow below graph indication (see **Figure 14**). Otherwise the pump may be damaged because the liquid is circulated within the pump, overheated because of full transfer of motor power into oil heating.

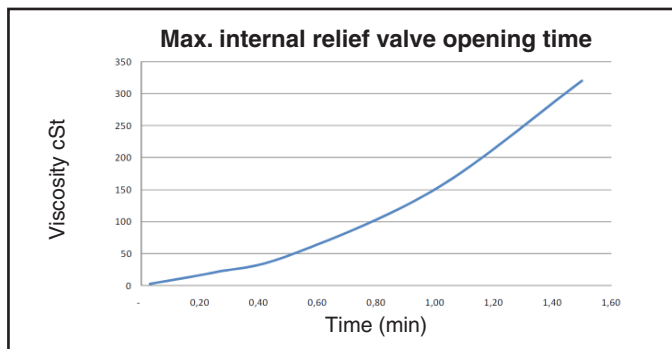


Figure 14



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4. Start Up, Operation & Shut Down (Continued)

4.12. Thermal Shock and Operating Temperature Limits

During pump start-up, as well as during pump operation, pump must not see a thermal shock greater than 28°C (50°F) from liquid entering the pump. Rapid temperature changes beyond this limit must be avoided.

Unless approved by Parker, liquids entering pump inlet must not be hotter than 107°C (225°F) nor colder than -18°C (0°F). Most pumps also have temperature limits of 107°C (225°F). The maximum rate of temperature change during pump heating or cooling should be about 0.8°C/minute (1.5°F/minute). A heated or cooled pump should be held at its start-up temperature for at least an hour prior to start-up. This will ensure uniform temperature distribution throughout pump assembly.



CAUTION/ATTENTION: Never exceed minimum or maximum allowable pump or liquid temperature. Do not expose equipment to thermal shock. Differences in metallurgy and their respective coefficients of expansion could cause distortion of pump parts resulting in a breakdown condition. Use of insulation and heating jacket or heat tracing to maintain pump at liquid temperature is recommended in high temperature applications.

4.13. Shutdown

If system is to be shut down for a short period, do not drain pump as this would require priming at start-up. If pump is to be stored, apply a rust-inhibiting agent (one compatible with all pump materials) to all internal and external surfaces, especially those that are machined.



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5. Maintenance



DANGER: BEFORE starting any maintenance procedure, do the following: Shut off all power switches and circuit breakers. Remove any electrical service fuses. Lock electrical service panel supplying power to driver. Shut, wire or chain, and lock all valves in pump inlet/outlet piping. If applicable, shut off any steam or other fluid supply lines to pump.

5.1. Filters and Strainers

All filter and strainer elements should be periodically checked for cleanliness and cleaned or replaced as necessary. This will protect equipment from damage due to pressure-drop across clogged or dirty elements.

5.2. Foundation

Foundation and hold-down bolts should be checked for tightness at least every six months.

5.3. Alignment

Alignment of pump and its driver should be checked and corrected, if necessary, at least every six months. If system experiences an unusual amount of vibrations or large variations in operating temperatures, this should be done often. Well-maintained alignment will help ensure maximum equipment life.



WARNING: Rotating parts, such as couplings, pulleys, external fans, or unused shaft extensions should be permanently guarded against accidental contact with personnel or clothing. This is particularly important where parts have surface irregularities such as keys or set- screws.

5.4. Lubrication

5.4.1. Bearings

Pump environment, operating conditions and intervals between bearing checks all effect bearing life.

Bearings have a finite life and should be checked often for increase in temperature and/or rough operation. If either condition is noted, stop equipment and replace bearing. When grease or oil fittings are provided, lubricate bearings as specified in applicable pump instruction manual.



CAUTION/ATTENTION: Continued running with a rough or worn bearing can lead to catastrophic bearing failure which could cause seal and/or pump failure.

5.5. Packing

A pump should be repacked when all packing gland travel is exhausted or when packing is damaged. Follow packing replacement instructions in applicable pump instruction manual.



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5. Maintenance (Continued)

5.6. Shaft Seals and Leakage

Visually check equipment frequently for signs of damage/leakage from shaft seals, gaskets or O-Rings. Be sure all connections are tight. If seal leakage is more than about 10 drops per hour per seal, shut down equipment and repair or replace necessary parts. Shaft seals have a finite life which is affected by operating conditions and environment. Expect them to wear and eventually fail. When leakage becomes unacceptable, replace seal unit with one compatible with pump's operating conditions. Dirty liquids will reduce seal life. Hollow Shaft pumps have connections for drain pipes.

NOTE: A very small amount of leakage (about 10 drops per hour per seal) is normal, even when the shaft seals are new.

NOTE: Make sure any seal leakage is disposed of properly.



WARNING: Since leakage or seal failure can be expected to eventually occur, be sure installation can withstand this situation. Take appropriate measures if liquid is hazardous.

5.7. Spare Parts

Where pump out-of-service time is of vital concern, and this down time must be minimized, a set of spare parts or repair kits should be retained on-site.

5.8. Disassembly and Reassembly

Various procedures for disassembly and reassembly apply to different pumps. Refer to the specific pump instruction manual on how to perform these procedures.



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6. Field and Factory Service & Parts

Parker maintains a staff of trained service personnel that can provide pump installation, pump start-up, maintenance/overhaul and troubleshooting supervision as well as installation and maintenance training.

Our factories provide maintenance as well as overhaul and test facilities in the event the user prefers to return pumps for inspection or overhaul. Pumps that have been factory-overhauled are normally tested and warranted “as-new” for a period of one year from date of shipment.

For either field service or factory overhaul assistance, contact your local Parker Sales Office or representative at the Technical/Customer Service Department in Parker (Gossolengo Piacenza) Italy. Most pumps have repair kits available. Minor Repair Kits are used to repair leaking seals, bad bearings and/or for re-assembly after pump tear-down. They include (as applicable) pump shaft seals, packing, all gaskets/O-rings and bearings. Major Repair Kits are sufficient to rebuild completely worn-out pumps to “as-new” condition. They include all parts found in Minor Repair Kits plus all major internal parts subject to wear. Since kits have all the necessary parts, it is preferred that they be purchased rather than selecting individual parts. When parts are individually selected from the Parts List, some needed components are often overlooked. In addition, mixing worn or used parts with new parts risks rapid wear and shortened service life from the new parts.

Please contact your local Parker Sales Office or representative at the Technical/Customer Service Department for spare part list of your pump model.



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7. Magnetic Coupling Driven Pump



DANGER: Be careful handling magnetic devices. Heart control systems, medical devices, electronics devices, components and tools may be affected and malfunctioning in case of usage near magnetic components, like magnetic joint used to connect parker screw pumps and motors. Injuries to personal health may happen.

7.1 What A Magnetic Coupling Driven Pump Looks Like

1. The shaft end is covered by a steel cover/cap.
2. The magnetic radiation exit through the steel cover.
3. Pay attention to personal injuries (See **Figure 15a**).

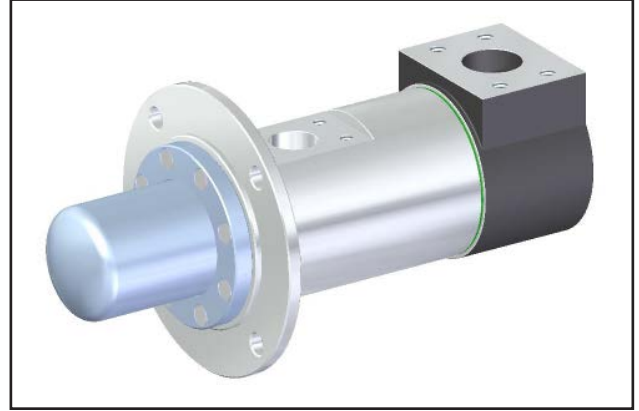
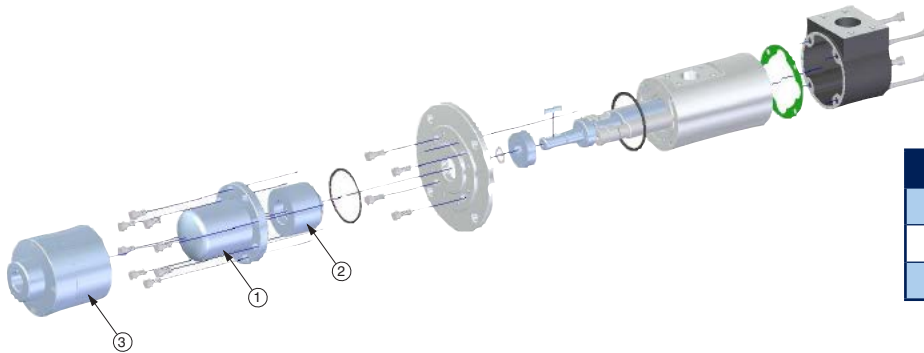
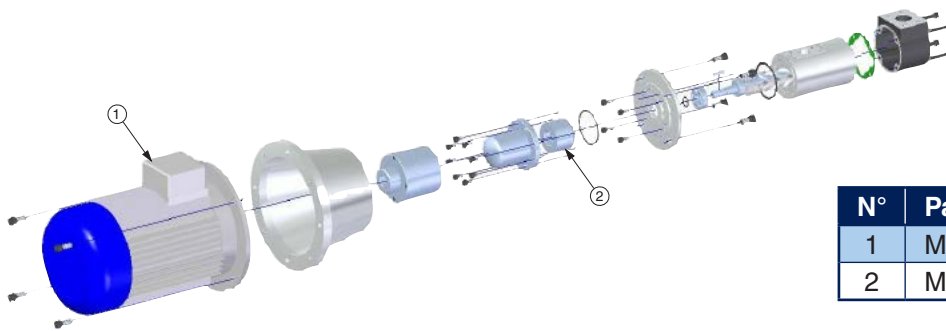


Figure 15a



N°	Parts Description
1	Containment shroud
2	Internal rotor
3	External rotor



N°	Parts Description
1	Magnetic coupling - motor part
2	Magnetic coupling - pump part



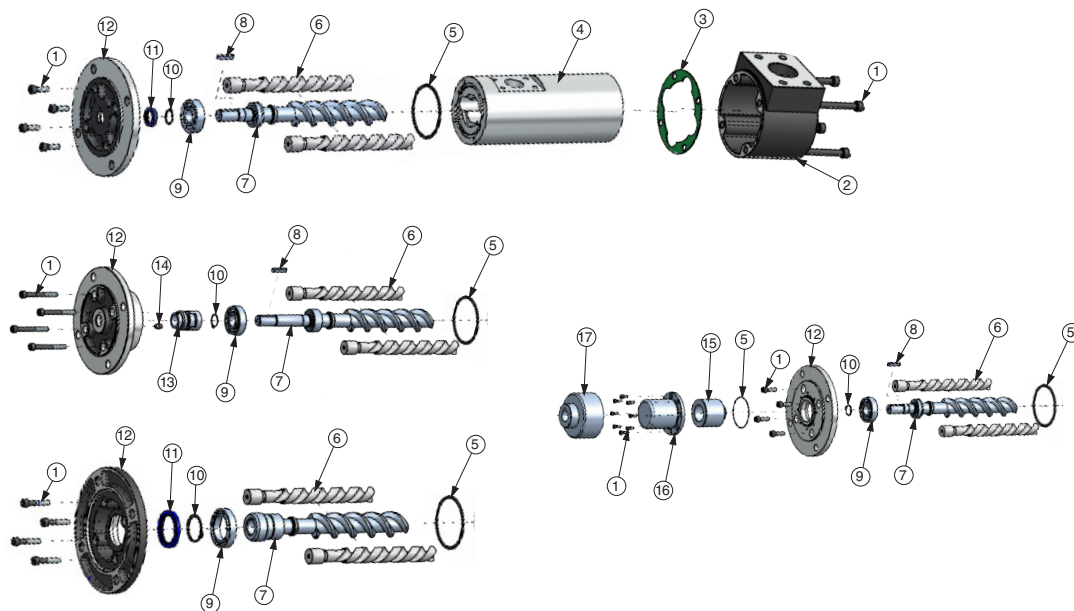
DANGER: Be Careful Handling Magnetic Devices. While assembling the motor pump unit, please pay attention to put near each other the part (A) and (B). An high magnetic force attract the components (A) and (B) and personal injuries may happen.



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8. Parts Description

SMT Pump Series – Front Configurations View

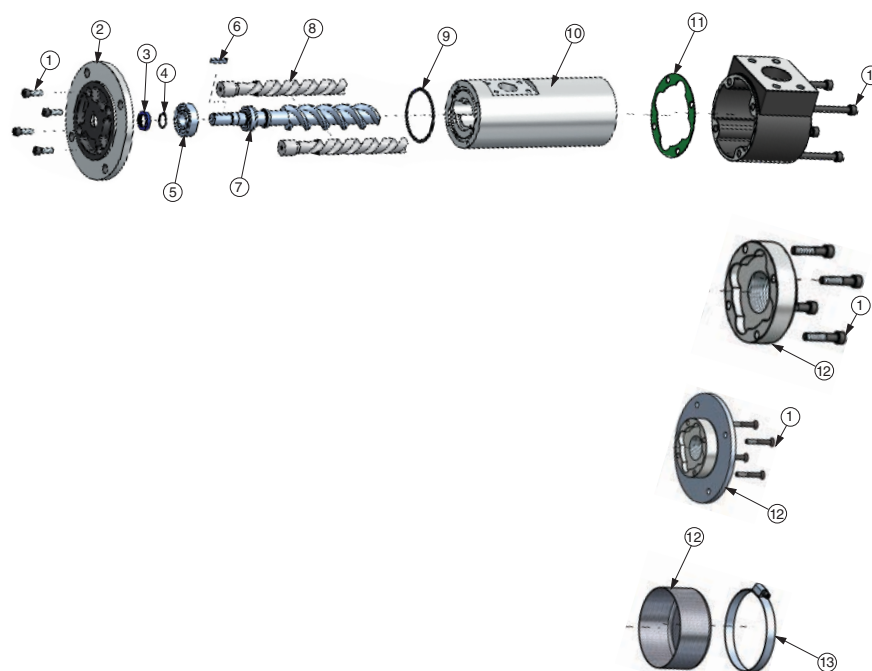


N°	Parts Description		
1	Capscrews		
2	Radial suction cover		
3	Plane gasket		
4	Pump body		
5	O-ring		
6	Idle screw		
N°	SMT	N°	SMT MECHANICAL SEAL
7	Main screw	7	Main screw
8	Key	8	Key
9	Key ball bearing	9	Key ball bearing
10	Seeger	10	Seeger
11	Seal	12	Flange
12	Flange	13	Mechanical seal
		14	Driving oil bushing
N°	SMT MAGNETIC COUPLING	N°	SMT HOLLOW SHAFT
7	Main screw	7	Main screw
8	Key	9	Key ball bearing
9	Key ball bearing	10	Seeger
10	Seeger	11	Seal
12	Flange	12	Flange
5	O-ring		
15	Internal motor MC		
16	Containment shroud MC		
17	External rotor MC		



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8. Parts Description (Continued)



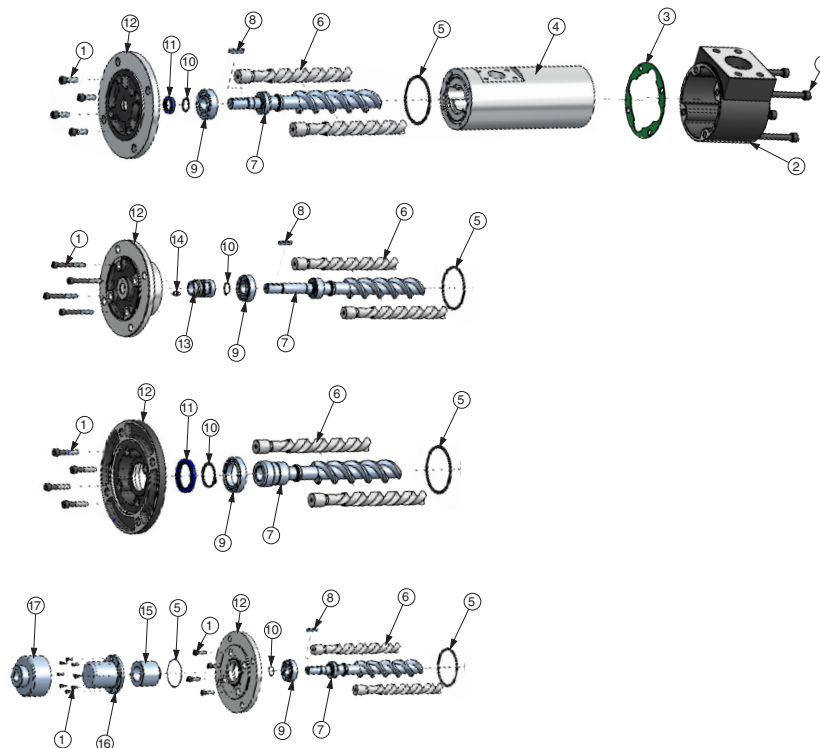
N°	Parts Description		
1	Capscrews		
2	Flange		
3	Seal		
4	Seeger		
5	Ball bearing		
6	Key		
7	Main screw		
8	Idler screw		
9	O-Ring		
10	Pump body		
N°	SMT	N°	SMT AXS (AXIAL SAE)
11	Plane gasket	11	Plane gasket
12	Radial suction cover	12	Radial suction cover
N°	SMT AL (TANK TOP FLANGE)	N°	SMT SUBMERSIBLE
11	Plane gasket	12	Filter
12	Radial suction cover	13	Clamp



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8. Parts Description (Continued)

SMT16B Pump Series – Front Configurations View



N°	Parts Description		
1	Capscrews		
2	Radial suction cover		
3	Plane gasket		
4	Pump body		
5	O-ring		
6	Idler screw		
N°	SMT16B	N°	SMT16B MECHANICAL SEAL
7	Main screw	7	Main screw
8	Key	8	Key
9	Ball bearing	9	Ball bearing
10	Seeger	10	Seeger
11	Seal	13	Mechanical seal
12	Flange	14	Driving oil bushing
		12	Flange
N°	SMT16B MAGNETIC COUPLING	N°	SMT16B HOLLOW SHAFT
7	Main screw	7	Main screw
9	Ball bearing	8	Ball bearing
10	Seeger	9	Seeger
12	Flange	10	Seal
15	O-ring	11	Flange
5	Internal motor MC		
16	Containment shroud MC		
17	External rotor MC		

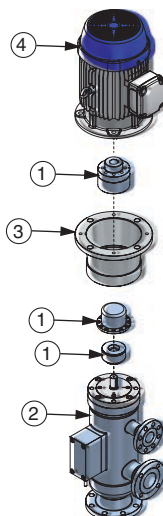


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8. Parts Description (Continued)

Vertical Assembly

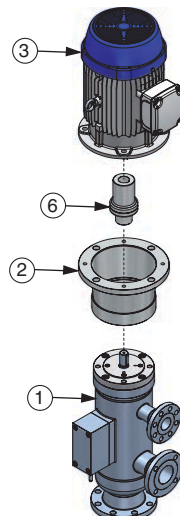
Motor Pump Group / Magnetic Coupling



N°	Parts Description
1	Magnetic coupling
2	SMAPI pump
3	Bell housing
4	Electric motor

Vertical Assembly

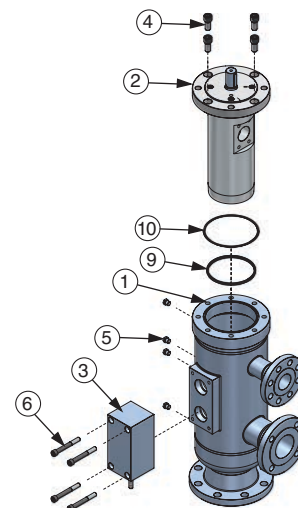
Motor Pump Group / Standard Coupling



N°	Parts Description
1	SMAPI pump
2	Bell housing
3	Electric motor
6	Elastic coupling

Vertical Assembly

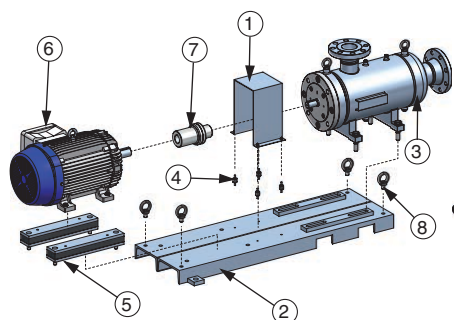
Pump



N°	Parts Description
1	Case V-ZN
2	Internal cartridge
3	RP external valve
4	Capscrews
5	Vent
6	Capscrews
9	Internal O-ring
10	Internal O-ring

Horizontal Assembly

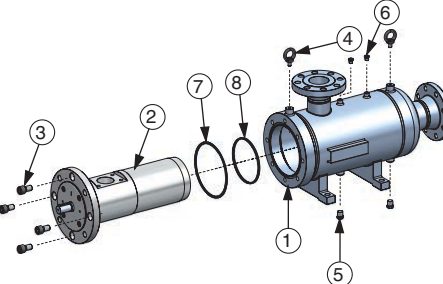
Motor Pump Group



N°	Parts Description
1	Carter
2	SKID
3	SMAPI pump
4	Vibration stopper
5	Shock absorber
6	Electric motor
7	Coupling
8	Golfare

Horizontal Assembly

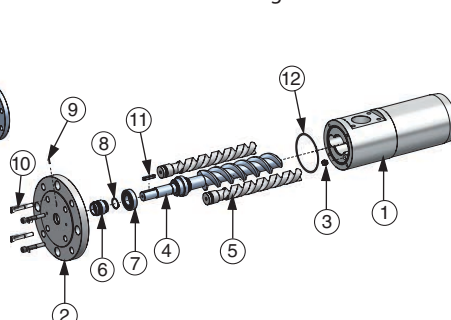
Pump



N°	Parts Description
1	External Case
2	Cartridge
3	Capscrews
4	Golfare
5	Drainer
6	Vent
7	Internal O-ring
8	Internal O-ring

Horizontal Assembly

Cartridge



N°	Parts Description
1	Pump housing
2	Flange
3	Bushing
4	Main screw
5	Idler screw
6	Mechanical seal
7	Ball bearing
8	Seeger
9	Pin
10	Capscrews
11	Key
12	Internal O-ring

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9. Troubleshooting

MALFUNCTION	POSSIBLE CAUSE	REPAIR
Loss of Flow or Low Capacity	System component malfunction	Inspect all system components Correct any malfunctions Ensure that suction and discharge lines are open and all valves are in proper positions
	Pump not primed or vented	Check reservoir oil level and fill as required Vent air from pump
	Low pump speed	Ensure driver is not overloaded For belt drives, ensure belt not slipping For variable speed drivers or variable speed intermediate devices, ensure proper speed is set
	Incorrect pump rotation	Correct direction of driver rotation
	Obstruction in piping	Inspect all system piping and valves Remove any obstructions
	Wear of rotors and/or housings	Replace worn rotors, gears, and/or housing(s)
	System bypass	Check all system bypass valves, including relief valve. Repair or replace as required
	Insufficient inlet pressure	Remove obstruction Clean suction strainer or filter element
The magnetic coupling is slipping	A too high viscosity has caused the magnetic coupling to slip	Lower the viscosity to a maximum approved and restart the pump by switching it off and on
	The pump is blocked inside the rotors by a foreign object	Remove the foreign object from the pump and control the bores and rotors for damages
Loss of Suction	Suction line closed, blocked or leaking	Verify suction line valve is locked open Inspect suction line, especially joints Remove any obstruction and repair any leaks Clean strainer or replace filter
	Excessive viscosity	Reduce viscosity by heating pump and/or system liquids
	Dirty suction strainer	Clean or replace strainer or filter element
	Wrong direction of rotation	Correct direction of driver rotation
Low Discharge Pressure	Low liquid level in the reservoir	Check liquid level in the reservoir and fill as necessary
	Air in system	Ensure pump is vented and suction lines are full of liquid
	Worn rotors , gears and/or housing(s)	Replace worn rotors, gears and/or housing(s)
	Obstruction in piping	Inspect inlet piping and suction valve(s) Remove any obstruction
	Dirty suction strainer	Clean/replace suction strainer or filter element
	System bypass problem	Check all system bypass valves for leakage, including relief valves Repair or replace as required
Excessive power usage	Fluid more viscous than specified	Heat fluid to proper viscosity and/or design temperature
	Pump suction and/or discharge lines closed or blocked	Ensure suction and discharge lines are open and remove obstructions if present
	Heavy internal rubbing of pump parts	Verify pump and driver alignment Inspect pump wearing parts and replace as necessary
	Excessive pump speed	Reduce pumps speed to design limits
	Mechanical problems	Check for bent shaft, tight shaft parking or pipe strain Repair or replace as required



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9. Troubleshooting (Continued)

MALFUNCTION	POSSIBLE CAUSE	REPAIR
Excessive or unusual noise or vibration	Misalignment	Check pump and driver and correct as required
	Restricted suction line	Check line and remove any obstructions
	Air in system	Ensure pump is vented and suction lines are full of fluid Check reservoir level and fill as necessary Check all lines, flanges, joints and connections for leakage. Repair as necessary
	Dirty suction strainer	Clean suction strainer or filter element
	Relief valve chatter or leakage	Check discharge relief valve pressure setting Re-adjust, repair or replace relief valve
	Heavy internal rubbing of pump parts	Verify pump and driver alignment Inspect pump wearing parts. Replace as necessary
	Mechanical problem	Check for loose or mispositioned coupling bent or broken shafts, or worn bearing. Repair or replace as required
Rapid pump wear	Fluid contains abrasive foreign matter	Collect samples of liquid and test for foreign matter Reduce downstream filter ratings in re-circulating systems (do not exceed NPIPR) If necessary, replace liquid
	Fluid contains water	Remove any water from the reservoir, find the source and prevent further contamination
	Misalignment	Check pump-driver alignment and correct as required
	Insufficient liquid	Check liquid level in the reservoir and correct as required Remove any suction line obstruction Clean/replace strainer or filter



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10. Application Description or Problem Reporting

Customer			
Project number or name			

APPLICATION/DESCRIPTION			
Power hydraulics			
Mobile			
Filtration			
Cooling			
Lubrication			
Elevator			
Other: please, specify			
Tank Capacity			
Cooling	YES ☐	NO ☐	
Installation	Above oil levels - Meters		Below oil levels - Meters

APPLICATION			
General Description			

PUMP TYPE			
Displacement (cc)			
Flow Rate		RPM for Above flow rate	

MOTOR SPEED of ROTATION			
Min. Speed		Max Speed	

OUTLET/OPERATING PRESSURE			
Max		Max. @ min. speed	

INLET/SUCTION PRESSURE			
Max		Max. @ min. speed	

FLUID			
Type/Designation			
Percentage of air		Percentage of water	

FLUID PROPERTIES @ STARTUP			
Viscosity [cSt]		Temperature [°C]	

FLUID PROPERTIES @ OPERATION			
Viscosity [cSt]		Temperature [°C]	



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This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There is no text or other markings on the paper.

Offer of Sale

1. Definitions. As used herein, the following terms have the meanings indicated.

- Buyer:** means any customer receiving a Quote for Products from Seller.
- Goods:** means any tangible part, system or component to be supplied by the Seller.
- Products:** means the Goods, Services and/or Software as described in a Quote provided by the Seller.
- Quote:** means the offer or proposal made by Seller to Buyer for the supply of Products.
- Seller:** means Parker Hannifin Corporation, including all divisions and businesses thereof.
- Services:** means any services to be supplied by the Seller.
- Software:** means any software related to the Products, whether embedded or separately downloaded.
- Terms:** means the terms and conditions of this Offer of Sale or any newer version of the same as published by Seller electronically at www.parker.com/saleterms.

2. Terms. All sales of Products by Seller are contingent upon, and will be governed by, these Terms and, these Terms are incorporated into any Quote provided by Seller to any Buyer. Buyer's order for any Products whether communicated to Seller verbally, in writing, by electronic data interface or other electronic commerce, shall constitute acceptance of these Terms. Seller objects to any contrary or additional terms or conditions of Buyer. Reference in Seller's order acknowledgement to Buyer's purchase order or purchase order number shall in no way constitute an acceptance of any of Buyer's terms of purchase. No modification to these Terms will be binding on Seller unless agreed to in writing and signed by an authorized representative of Seller.

3. Price; Payment. The Products set forth in Seller's Quote are offered for sale at the prices indicated in Seller's Quote. Unless otherwise specifically stated in Seller's Quote, prices are valid for thirty (30) days and do not include any sales, use, or other taxes or duties. Seller reserves the right to modify prices at any time to adjust for any raw material price fluctuations. Unless otherwise specified by Seller, all prices are F.C.A. Seller's facility (INCOTERMS2010). All sales are contingent upon credit approval and payment for all purchases is due thirty (30) days from the date of invoice (or such date as may be specified in the Quote). Unpaid invoices beyond the specified payment date incur interest at the rate of 1.5% per month or the maximum allowable rate under applicable law.

4. Shipment; Delivery: Title and Risk of Loss. All delivery dates are approximate. Seller is not responsible for damages resulting from any delay. Regardless of the manner of shipment, delivery occurs and title and risk of loss or damage pass to Buyer, upon placement of the Products with the shipment carrier at Seller's facility. Unless otherwise agreed, Seller may exercise its judgment in choosing the carrier and means of delivery. No deferment of shipment at Buyer's request beyond the respective indicated shipping date will be made except on terms that will indemnify, defend and hold Seller harmless against all loss and additional expense. Buyer shall be responsible for any additional shipping charges incurred by Seller due to Buyer's acts or omissions.

5. Warranty. The warranty related to the Products is as follows: (i) Goods are warranted against defects in material or workmanship for a period of twelve (12) months from the date of delivery or 2,000 hours of use, whichever occurs first; (ii) Services shall be performed in accordance with generally accepted practices and using the degree of care and skill that is ordinarily exercised and customary in the field to which the Services pertain and are warranted for a period of six (6) months from the completion of the Services by Seller; and (iii) Software is only warranted to perform in accordance with applicable specifications provided by Seller to Buyer for ninety (90) days from the date of delivery or, when downloaded by a Buyer or end user, from the date of the initial download. All prices are based upon the exclusive limited warranty stated above, and upon the following disclaimer: **DISCLAIMER OF WARRANTY: THIS WARRANTY IS THE SOLE AND ENTIRE WARRANTY PERTAINING TO PRODUCTS. SELLER DISCLAIMS ALL OTHER WARRANTIES, EXPRESS AND IMPLIED, INCLUDING DESIGN, NON INFRINGEMENT, MERCHANTABILITY, AND FITNESS FOR A PARTICULAR PURPOSE. SELLER DOES NOT WARRANT THAT THE SOFTWARE IS ERROR-FREE OR**

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6. Claims; Commencement of Actions. Buyer shall promptly inspect all Products upon receipt. No claims for shortages will be allowed unless reported to the Seller within ten (10) days of delivery. Buyer shall notify Seller of any alleged breach of warranty within thirty (30) days after the date the non-conformance is or should have been discovered by Buyer. Any claim or action against Seller based upon breach of contract or any other theory, including tort, negligence, or otherwise must be commenced within twelve (12) months from the date of the alleged breach or other alleged event, without regard to the date of discovery.

7. LIMITATION OF LIABILITY. IN THE EVENT OF A BREACH OF WARRANTY, SELLER WILL, AT ITS OPTION, REPAIR OR REPLACE THE NON-CONFORMING PRODUCT, RE-PERFORM THE SERVICES, OR REFUND THE PURCHASE PRICE PAID WITHIN A REASONABLE PERIOD OF TIME. IN NO EVENT IS SELLER LIABLE FOR ANY SPECIAL, INDIRECT, INCIDENTAL OR CONSEQUENTIAL DAMAGES ARISING OUT OF, OR AS THE RESULT OF, THE SALE, DELIVERY, NON-DELIVERY, SERVICING, NON-COMPLETION OF SERVICES, USE, LOSS OF USE OF, OR INABILITY TO USE THE PRODUCTS OR ANY PART THERE OF, LOSS OF DATA, IDENTITY, PRIVACY, OR CONFIDENTIALITY, OR FOR ANY CHARGES OR EXPENSES OF ANY NATURE INCURRED WITHOUT SELLER'S WRITTEN CONSENT, WHETHER BASED IN CONTRACT, TORT OR OTHER LEGAL THEORY. IN NO EVENT SHALL SELLER'S LIABILITY UNDER ANY CLAIM MADE BY BUYER EXCEED THE PURCHASE PRICE PAID FOR THE PRODUCTS.

8. Loss to Buyer's Property. Any designs, tools, patterns, materials, drawings, confidential information or equipment furnished by Buyer or any other items which are or become Buyer's property, will be considered obsolete and maybe destroyed by Seller after two (2) consecutive years have elapsed without Buyer ordering the Products manufactured using such property. Seller shall not be responsible for any loss or damage to such property while it is in Seller's possession or control.

9. Special Tooling. Special Tooling includes but is not limited to tooling, jigs, fixtures and associated manufacturing equipment acquired or necessary to manufacture Products. A tooling charge may be imposed for any Special Tooling. Such Special Tooling shall be and remain Seller's property not withstanding payment of any charges by Buyer. In no event will Buyer acquire any interest in Special Tooling belonging to Seller that is utilized in the manufacture of the Products, even if such Special Tooling has been specially converted or adapted for such manufacture and not withstanding any charges paid by Buyer. Unless otherwise agreed, Seller has the right to alter, discard or otherwise dispose of any Special Tooling or other property in its sole discretion at any time.

10. Security Interest. To secure payment of all sums due, Seller retains a security interest in all Products delivered to Buyer and, Buyer's acceptance of these Terms is deemed to be a Security Agreement under the Uniform Commercial Code. Buyer authorizes Seller as its attorney to execute and file on Buyer's behalf all documents Seller deems necessary to perfect its security interest.

11. User Responsibility. The Buyer through its own analysis and testing, is solely responsible for making the final selection of the Products and assuring that all performance, endurance, maintenance, safety and warning requirements of the application of the Products are met. The Buyer must analyze all aspects of the application and follow applicable industry standards, specifications, and other technical information provided with the Product. If Seller provides Product options based upon data or specifications provided by the Buyer, the Buyer is responsible for determining that such data and specifications are suitable and sufficient for all applications and reasonably foreseeable uses of the Products. In the event the Buyer is not the end-user, Buyer will ensure such end-user complies with this paragraph.

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12. Use of Products, Indemnity by Buyer. Buyer shall comply with all instructions, guides and specifications provided by Seller with the Products. Unauthorized Uses. If Buyer uses or resells the Products for any uses prohibited in Seller's instructions, guides or specifications, or Buyer otherwise fails to comply with Seller's instructions, guides and specifications, Buyer acknowledges that any such use, resale, or non-compliance is at Buyer's sole risk. Buyer shall indemnify, defend, and hold Seller harmless from any losses, claims, liabilities, damages, lawsuits, judgments and costs (including attorney fees and defense costs), whether for personal injury, property damage, intellectual property infringement or any other claim, brought by or incurred by Buyer, Buyer's employees, or any other person, arising out of: (a) improper selection, application, design, specification or other misuse of Products provided by Seller; (b) any act or omission, negligent or otherwise, of Buyer; (c) Seller's use of patterns, tooling, equipment, plans, drawings, designs or specifications or other information or things furnished by Buyer; (d) damage to the Products from an external cause, repair or attempted repair by anyone other than Seller, failure to follow instructions, guides and specifications provided by Seller, use with goods not provided by Seller, or opening, modifying, deconstructing or tampering with the Products for any reason; or (e) Buyer's failure to comply with these Terms. Seller shall not indemnify Buyer under any circumstance except as otherwise provided in these Terms.

13. Cancellations and Changes. Buyer may not cancel or modify any order for any reason, except with Seller's written consent and upon terms that will indemnify, defend and hold Seller harmless against all direct, incidental and consequential loss or damage. Seller, at any time, may change Product features, specifications, designs and availability.

14. Limitation on Assignment. Buyer may not assign its rights or obligations without the prior written consent of Seller.

15. Force Majeure. Seller does not assume the risk and is not liable for delay or failure to perform any of Seller's obligations by reason of events or circumstances beyond its reasonable control ("Events of Force Majeure"). Events of Force Majeure shall include without limitation: accidents, strikes or labor disputes, acts of any government or government agency, acts of nature, delays or failures in delivery from carriers or suppliers, shortages of materials, or any other cause beyond Seller's reasonable control.

16. Waiver and Severability. Failure to enforce any provision of these Terms will not invalidate that provision; nor will any such failure prejudice Seller's right to enforce that provision in the future. Invalidation of any provision of these Terms by legislation or other rule of law shall not invalidate any other provision herein and, the remaining provisions will remain in full force and effect.

17. Termination. Seller may terminate any agreement governed by or arising from these Terms for any reason and at anytime by giving Buyer thirty (30) days prior written notice. Seller may immediately terminate, in writing, if Buyer: (a) breaches any provision of these Terms (b) appoints a trustee, receiver or custodian for all or any part of Buyer's property (c) files a petition for relief in bankruptcy on its own behalf, or one if filed by a third party (d) makes an assignment for the benefit of creditors; or (e) dissolves its business or liquidates all or a majority of its assets.

18. Ownership of Software. Seller retains ownership of all Software supplied to Buyer hereunder. In no event shall Buyer obtain any greater right in and to the Software than a right in the nature of a license limited to the use thereof and subject to compliance with any other terms provided with the Software.

19. Indemnity for Infringement of Intellectual Property Rights. Seller is not liable for infringement of any patents, trademarks, copyrights, trade dress, trade secrets or similar rights ("Intellectual Property Rights") except as provided in this Section. Seller will defend at its expense and will pay the cost of any settlement or damages awarded in an action brought against Buyer based on a third party claim that one or more of the Products sold hereunder infringes the Intellectual Property Rights of a third party in the country of delivery of the Products by the Seller to the Buyer. Seller's obligation to defend and indemnify Buyer is contingent on Buyer notifying Seller within ten (10) days after Buyer becomes aware of any such claim, and Seller having sole control over the defense of the claim including all negotiations for settlement or compromise. If one or more Products sold hereunder is subject to "such a claim,

Seller may, at its sole expense and option, procure for Buyer the right to continue using the Products, replace or modify the Products so as to render them non-infringing, or offer to accept return of the Products and refund the purchase price less a reasonable allowance for depreciation. Seller has no obligation or liability for any claim of infringement: (i) a rising from information provided by Buyer; or (ii) directed to any Products provided hereunder for which the designs are specified in whole or part by Buyer; or (iii) resulting from the modification, combination or use in a system of any Products provided hereunder. The foregoing provisions of this Section constitute Seller's sole and exclusive liability and Buyer's sole and exclusive remedy for such claims of infringement of Intellectual Property Rights.

20. Governing Law. These Terms and the sale and delivery of all Products are deemed to have taken place in, and shall be governed and construed in accordance with, the laws of the State of Ohio, as applicable to contracts executed and wholly performed therein and without regard to conflicts of laws principles. Buyer irrevocably agrees and consents to the exclusive jurisdiction and venue of the courts of Cuyahoga County, Ohio with respect to any dispute, controversy or claim arising out of or relating to the sale and delivery of the Products.

21. Entire Agreement. These Terms, along with the terms set forth in the main body of any Quote, forms the entire agreement between the Buyer and Seller and constitutes the final, complete and exclusive expression of the terms of sale. In the event of a conflict between any term set forth in the main body of a Quote and these Terms, the terms set forth in the main body of the Quote shall prevail. All prior or contemporaneous written or oral agreements or negotiations with respect to the subject matter shall have no effect. These Terms may not be modified unless in writing and signed by an authorized representative of Seller.

22. Compliance with Laws. Buyer agrees to comply with all applicable laws, regulations, and industry and professional standards, including those of the United States of America, and the country or countries in which Buyer may operate, including without limitation the U.S. Foreign Corrupt Practices Act ("FCPA"), the U.S. Anti-Kickback Act ("Anti-Kickback Act"), U.S. and E.U. export control and sanctions laws ("Export Laws"), the U.S. Food Drug and Cosmetic Act ("FDCA"), and the rules and regulations promulgated by the U.S. Food and Drug Administration ("FDA"), each as currently amended. Buyer agrees to indemnify, defend, and hold harmless Seller from the consequences of any violation of such laws, regulations and standards by Buyer, its employees or agents. Buyer acknowledges that it is familiar with all applicable provisions of the FCPA, the Anti-Kickback Act, Export Laws, the FDCA and the FDA and certifies that Buyer will adhere to the requirements thereof and not take any action that would make Seller violate such requirements. Buyer represents and agrees that Buyer will not make any payment or give anything of value, directly or indirectly, to any governmental official, foreign political party or official thereof, candidate for foreign political office, or commercial entity or person, for any improper purpose, including the purpose of influencing such person to purchase Products or otherwise benefit the business of Seller. Buyer further represents and agrees that it will not receive, use, service, transfer or ship any Product from Seller in a manner or for a purpose that violates Export Laws or would cause Seller to be in violation of Export Laws."

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